

Aditya Tayal

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Final-year CSE undergraduate at IIT Mandi who ships software people actually use. Solo-built PlanMyDegree, the degree planner **1,741 IIT Mandi students** run their semesters on, plus exam, leave and lab systems for the institute. Four startup internships, an **All India Rank 1** in an IEEE ICASSP challenge, 20+ hackathons, and the elected **UG Academic Affairs Secretary** for ~3,500 undergraduates.

EDUCATION

Indian Institute of Technology Mandi

B.Tech in Computer Science & Engineering | CGPA: 8.80

Mandi, Himachal Pradesh

Aug 2023 – May 2027 (expected)

Relevant Coursework: Machine Learning, Deep Learning, Generative AI, Data Science I & II, Data Handling & Visualization, Database Systems, Design of Algorithms, Automata Theory, Operating Systems, Computer Networks, Information Security & Privacy, Optimization

Ramakrishna Mission

Higher Secondary Education in PCM with Computer Science | 89.40%

Gwalior, Madhya Pradesh

Aug 2020 – Jun 2022

Wendy School Junior College

ICSE Secondary Education | 98.20%

Gwalior, Madhya Pradesh

Apr 2007 – Jul 2020

EXPERIENCE

Major Technical Project, American Express

Imbalanced fraud detection | Python, scikit-learn, XGBoost, LightGBM, CatBoost

Aug 2026 – Present

Team project

- Stress-testing Self-Paced Ensemble learning on a multi-product fraud portfolio built from public data: 4 base learners, pooled vs per-product, 3 seeds, scored by a cost model that turns fraud ranking into money saved; the best setup recovers **64%** of what is recoverable.

Software Developer Intern, Koha Consultants (InstaFarms & MagoStays)

Travel tech | TypeScript, Node.js, PostgreSQL, Next.js, React Native

Apr 2026 – Present

- Full-stack developer on a two-brand villa and farmstay booking platform, owning features across payments, invoicing and bookings in the backend, admin panel, customer sites and host app, with a bias for correctness, audit trails and tests.

AI/ML Engineer Intern, WattCycle

Solar energy | Python, FastAPI, scikit-learn, XGBoost, Prophet, PostgreSQL, React

Dec 2025 – Jun 2026

- Worked across ML and full stack on WattCycle Atlas, a solar-plant monitoring platform: fault detection, energy forecasting and the analytics backend behind its dashboards.

Full Stack Developer Intern, Garudakshak Technologies

Defence tech | Tauri, Go, React, TypeScript, PostgreSQL

Oct 2025 – Apr 2026

- Built the operator desktop application for a counter-drone system from an empty repository to MVP, shipped for Windows and Raspberry Pi.

Lead Developer Intern, Strive Partners

Hiring tech | React, TypeScript, JavaScript, Django REST, AWS

Sep 2025 – Jun 2026

- Led development of Strive Hunt, a hiring platform with candidate and employer portals and an internal applicant-tracking system.

PROJECTS

Built for IIT Mandi

PlanMyDegree: the degree planner IIT Mandi runs on (Solo)

Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, PWA

Jan 2026 – Present

- **1,741 students** plan credits, minors, timetables and pre-registration on it; it took **150+ sign-ups in 36 hours** from one WhatsApp message and no ads.

Faculty Leave Portal (Solo)

Fastify, Prisma, React, SQLite, Docker

Sep 2026 – Present

- Leave management that never stores a balance: every figure is derived from an append-only ledger the database refuses to edit, with **324 requirements** traced to **321 tests**.

TestPlatform: online examinations (Solo)

Next.js, Express, Socket.IO, PostgreSQL, Redis

Aug 2026 – Present

- An exam system where the clock lives on the server, not the browser: Redis-held timers, live rooms, a differently shuffled paper per student, auto-grading and passwordless OTP.

Exodia 2026 Ticketing & Gate (Solo)

Next.js, tRPC, Prisma, PostgreSQL, Nodemailer

[Code](#)

Mar 2026 – Apr 2026

- Ran entry for the fest’s biggest night: **300+ QR tickets**, **200 entries**, and **26 reuse attempts blocked** by single-use database claims, with **13 commits** shipped live on the day.

IIT Mandi LaTeX (Lead)

Overleaf CE, Node.js, Docker

Sep 2026 – Present

- An institute-hosted Overleaf for ~3,500 users, extended with a first-come compile queue, a 50-user cap with a waiting room, per-user quotas and nightly backups.

TinkerBench & Simulation Lab (Solo)

JavaScript, Canvas, nginx

Sep 2026 – Oct 2026

- A browser electronics bench that solves the circuit by nodal analysis every frame and runs Arduino code, plus **49** interactive physics and chemistry experiments for school labs.

Faculty Application Portal

Django, SQL

Dec 2024 – Feb 2025

- Replaced Google Forms with a Django portal on IIT Mandi’s recruitment site, with an admin dashboard to filter and export applicants and automated CVs and emails.

Machine Learning & Research

Camouflaged Object Detection (Paid Research)

PyTorch, mmdetection, DINOv2, LoRA, SAM2

Apr 2026 – Jul 2026

- Finding objects built not to be found: adapted PAIS to 10% labels (**+7.8 AP**), designed a text-guided DINOv2 + LoRA decoder (S-measure **0.847**), and edged past the UCOD-DPL paper (**0.8407 vs 0.835**).

Unlearning a Domain in CLIP (Team of 4)

PyTorch, CLIP ViT-B/16, prompt learning

Jan 2026 – May 2026

- Made CLIP forget a whole visual domain while keeping the rest: beat the published method (**72.10 vs 69.96 H-score**), then hit **83.6** by adapting SCRUB to domains, **+13.6** over the paper.

ALS Severity from Voice, SAND 2025 (Team of 4)

Python, audio ML

Oct 2025 – Nov 2025

- Graded ALS severity from speech recordings for an IEEE ICASSP challenge: **All India Rank 1, Global Rank 11** (macro-F1 0.5430); India #2 on the progression task.

Vehicle Counting with Deep Learning Banned (Solo)

Python, OpenCV, NumPy

Jan 2026

- Counted traffic with classical computer vision alone: a 9-stage pipeline built in 48 hours that finished **rank 10 against teams with MAE 16.4**.

MahaDrishti AI: Satellite Crop Verification (Team of 3)

Sentinel-1/2, Prithvi-EO 2.0, Random Forest, FastAPI

May 2026

- Checks a farmer’s declared crop against satellite imagery plot by plot and writes the report in Marathi; **finalist** at the Pune International Agri Hackathon.

Product Price Prediction, Amazon ML Challenge (Team of 3)

LightGBM, TF-IDF

Oct 2025

- Priced 75,000 products from their text alone with a 21-model LightGBM ensemble, cutting SMAPE from **54.38% to 32.33%**.

Deepfake Detection (Team of 6)

CLIP, DINOv2, SVM (cuML)

Feb 2025 – May 2025

- Caught images from **19 different generators** with frozen vision-transformer features and a GPU SVM: **89.4%** accuracy across 80,353 test images.

CaptureSmart: AI Camera Assistant (Team of 6)

Kotlin, CameraX, TensorFlow, Flask

Code

May 2025

- An Android camera that scores blur live (MobileNetV2, Pearson **0.89** on 24,000 images) and suggests ISO and shutter speed; **3rd of 60+ teams**.

Applied AI & Hackathons

FinFabric, IDBI Innovate 2026 (Team Lead)

JavaScript, scoring models, Vercel

[Code](#)

Jul 2026

- Led five live banking prototypes in one suite: MSME health scoring, a mutual-fund SIP companion, lead scoring, default early warning and fuzzy-matching reconciliation.

Sahayak, KSP Datathon 2026 (Solo)

Qwen 2.5 Coder, FastAPI, React

[Code](#)

Jun 2026

- Ask a police crime database a question in English or Kannada and get back the SQL, the rows behind the answer and the matching legal codes, all behind a read-only guard.

SevaVoice, HACKHAZARDS '26 (Solo)

React, TypeScript, Express, Sarvam AI

[Live Code](#)

Jun 2026

- Turns confusing public-service notices into step-by-step plans in **14 Indian languages**, read aloud in 11.

DeepMatch, India Runs Hackathon (Solo)

Python, sentence-transformers

[Code](#)

Jun 2026

- Ranks 100K candidates down to a shortlist in **~75 s on a CPU**, with six heuristics that catch fake profiles.

DyslexoFly, Code for Bharat S2 (Team of 2)

Next.js, Flask, DistilBART, mT5, edge-tts

[Live Code](#)

Jun 2025 – Jul 2025

- Turns PDFs into dyslexia-friendly reading: OpenDyslexic type, English and Hindi summaries, and audio in 7 neural voices.

Adobe India Hackathon 2025 (Team Leader)

Python, PyMuPDF, MiniLM, Docker

[1A](#) [1B](#)

Jul 2025

- **Top 1.5%** of 262,120 teams with multilingual PDF outline extraction in an offline, CPU-only container and persona-aware section ranking.

More projects

- **GBNet improvements** (research): beat the reproduced baseline on all **42/42** dataset-metric checks over 71 documented experiment phases.
- **Face EngagementCV** [Code](#): names every face in classroom CCTV and scores engagement from brightness, eye openness and motion, on a CPU.
- **PSCDL 2026** [Code](#): persistent scene-change detection with classical CV, two persistence clocks and an adaptive threshold.
- **Vidyut AI**, ET AI Hackathon 2.0 [Code](#): industrial-safety answers with citations to OSHA, CSB and Indian standards, in English and Hindi.
- **FlowAI**, Vibe2Ship 2026 [Code](#): a Gemini multi-agent planner that splits goals into subtasks, schedules them and nudges you.
- **PurrPedia**, HackTheKitty 2026 [Code](#): a 247-breed cat encyclopedia on Temporal workflows that passed its security audit after 5 fixes.
- **CareerAutofill AI** [Code](#): agents plus a Chrome extension that fill job applications; LoRA cut perplexity from 35.7 to 23.2.
- **TheBudgetBuddy**, FrostHack 2025 [Code](#): plain-English questions over your own transactions, via a LangChain agent on Llama3-70B with FAISS.
- **SecureInvest-AI**, SEBI Securities Market Hackathon [Code](#): a fraud-risk scorer for investment pitches, built solo in 48 hours.
- **Malware scanner**, KrackHack [Code](#): static scanning of .exe, .pdf and .docx files with YARA rules and PE import analysis.
- **TheAttendioBot** [Code](#): a Telegram bot that tells you how many classes you can still skip, with rate-limit auto-blocking.
- **Neufin Live** [Code](#): a product catalogue with search, pagination and a cart, validated with Zod and tested with Jest.
- **CyberShield**, PSBs Hackathon at IIT Hyderabad: mule-account detection on 9,082 accounts at a 0.89% base rate.
- **PodSense**, ABB Accelerator 2026: designed a four-agent Kubernetes pod observability system with an orchestrator.
- **Solar-panel waste policy study** (team of 5): tested whether India's battery-recycling rules can carry over to solar waste, with a field visit.
- **PRAYAS ML labs**: four teaching notebooks, from decision trees to DQN and TFLite, with 60 guided experiments.
- **Campus self-hosting**: six campus services on one Raspberry Pi 5, reachable through redirects that follow its changing address.
- **Research-group website**: an Astro site whose admin panel rebuilds and commits itself, with undo.
- **Autonomous ROS robot** (team of 5): an ESP32 on a custom 2-layer PCB, steered by a vector-based ROS planner.
- Also: HackerRank Orchestrate entry, the GCS cell website, and LaTeX study notes for Data Structures, DBMS and Python.

POSITIONS OF RESPONSIBILITY

UG Academic Affairs Secretary, Student Gymkhana, IIT Mandi

Jul 2026 – Present

- Elected student lead for academics for **~3,500 undergraduates**; head the UG Academic Council, picked **12 branch councillors** and ran Batch 2026's orientation across **15 branch sessions**.

Batch 2023 Academic Representative & Academic Guide

2024 – 2026

- The link between **500+ students** and faculty; co-ran the first post-COVID offline curriculum session and hired the council's web team (8 candidates, 4 selected).

Head of Technical Team, Exodia 2026 (Annual Cultural Fest)

Apr 2026

- Ran the fest's tech, from ticketing and the gate to the website's performance.

Instructor, HIMSHIKHAR (CCE IIT Mandi × Masai) & PRAYAS 4.0

Jun 2026 – Jul 2026

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- Taught **22 classes (40+ hours)** from DBMS to GenAI and an 18-hour ML course to learners from Class 9 to over 80, leading **8 TAs**.

UG Head Teaching Assistant, Machine Learning

- Ran course operations for **600+ students** across a semester.

Mentoring & Clubs

2023 – Present

- GCS Cell Mentor, SAE Club Mentor, STAC Core Member (hackathons for **100+ teams**), JEE Mentor at Physics Wallah (**100+** aspirants).

ACHIEVEMENTS

AIR 1, Global Rank 11, SAND 2025 (IEEE ICASSP 2026 challenge): ALS severity from voice; India #2, Global 19 in Task 2.

Rank 10, solo vs teams, Vehant Vehicle Count Challenge: classical CV only, MAE 16.4 (Jan 2026).

Finalist, Pune International Agri Hackathon 2026: Geospatial Crop Survey track (May 2026).

3rd Rank, AiHack India Hackathon 2025: hosted at IIT Mandi by Aiful Corporation, Japan (Nov 2025).

Round 2 (top 1.5%), Adobe India Hackathon 2025: team leader among 262,120 teams (Jul 2025).

Emerald of Gymkhana Award: best GCS Cell volunteer, Dean Students' Office (May 2025).

National Semi-Finalist, Tata Imagination Challenge 2024: through 2 national rounds (Feb 2025).

1st Rank, Space Code Challenge 2023: STAC, IIT Mandi (Jan 2024).

20+ hackathons & data challenges in 2025–26, incl. Amazon ML Challenge 2025 (82,787 teams), IDBI Innovate 2026 (team lead), ABB Accelerator 2026, SEBI GFF 2025.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C, C++, Java, Kotlin, SQL, Rust (Tauri), R, HTML/CSS, LaTeX

Backend: FastAPI, Flask, Django, Node.js, Express, Fastify, Hono, tRPC, REST, WebSockets, Socket.IO

Frontend & Apps: React, Next.js, Vite, Tailwind, Astro, Tauri, React Native (Expo), Android (Kotlin), PWAs

Data: PostgreSQL, SQLite, MongoDB, Redis, Prisma, Drizzle, Firestore, FAISS

ML & AI: PyTorch, TensorFlow/Keras, scikit-learn, LightGBM, XGBoost, CatBoost, OpenCV, Transformers, LoRA/PEFT, CLIP, DINOv2, SAM2, LangChain, RAG

Cloud & DevOps: Docker, AWS, DigitalOcean, Vercel, Render, Firebase, nginx, pm2, CI/CD, Playwright, Linux, Raspberry Pi